

A Hybrid Multi-Agent and RAG Architecture for Simulating Information-Rich Financial Markets

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Abstract—Agent-based models have long been used to simulate financial markets, yet most rely on agents with no access to external information, limiting the realism of emergent market dynamics. This paper introduces a hybrid architecture that combines multi-agent simulation with Retrieval-Augmented Generation, enabling trading agents to retrieve relevant financial news from a corpus of 23,515 items and reason about it through a large language model before executing trades. Four agent types operate within the simulation: fundamentalist, sentiment-momentum, noise, and RAG-enhanced traders. Three experimental conditions were tested across nine runs: a baseline with no information access, a random news control where agents received irrelevant text, and a full RAG condition with semantically relevant retrieval. Results were evaluated against established stylised facts of real financial markets, including excess kurtosis, volatility clustering, and non-stationarity. The random news condition produced the most realistic return distribution, achieving an excess kurtosis of 3.46 compared to 1.95 for full RAG and 1.05 for baseline. Analysis of agent behaviour revealed that RAG-informed agents defaulted to holding positions in 80.5% of steps due to confidence-based filtering, effectively suppressing the volatility needed for realistic fat-tailed returns. These findings highlight a tension between agent rationality and market realism, suggesting that behavioural noise may be as important as informed reasoning in producing realistic simulated markets.

I. INTRODUCTION

Financial markets are complex adaptive systems shaped by the decisions of millions of participants, each acting on different information, beliefs, and strategies. Agent-based models (ABMs) have become a widely used tool for studying how these individual behaviours give rise to aggregate market phenomena [1], [2]. Early work such as the Santa Fe Artificial Stock Market showed that even simple heterogeneous agents could reproduce patterns seen in real price data [2], while Cont’s landmark study established a set of statistical regularities known as stylised facts that any credible market model should exhibit [3]. These include fat-tailed return distributions, volatility clustering, and the absence of linear autocorrelation in returns.

Despite decades of progress in ABM finance [1], [4], one aspect of real trading remains poorly captured: the role of information. Real market participants constantly consume news, interpret earnings reports, and react to macroeconomic signals. Yet most simulated agents operate in an information vacuum. They follow fixed rules or respond to price alone, with no mechanism for reading, interpreting, or reasoning

about external data. This disconnect limits how realistic their collective behaviour can be.

Recent advances in large language models (LLMs) have opened a path toward closing this gap. Park et al. demonstrated that LLM-powered agents can exhibit believable, context-sensitive behaviour in simulated social environments [5]. In parallel, Retrieval-Augmented Generation (RAG) has emerged as a practical method for grounding LLM outputs in factual, retrieved content rather than relying solely on parametric memory [6]. RAG has already been applied to financial question answering [7], and open-source financial LLMs such as FinGPT have expanded the toolkit available for sentiment-aware applications [8]. However, no existing work has integrated RAG-enhanced agents into a multi-agent financial market simulation and evaluated the resulting dynamics against established stylised facts.

This paper addresses that gap directly. We propose a hybrid architecture in which trading agents retrieve semantically relevant financial news from a corpus of 23,515 items using FAISS and sentence-level embeddings [9], then pass the retrieved context to Google’s Gemini 2.5 Flash-Lite model for reasoning. The model returns a structured JSON trading decision that includes a confidence score; agents with low confidence default to holding their position. Four distinct agent types populate the simulation: fundamentalist, sentiment-momentum, noise, and RAG-enhanced running within the Mesa 3.x framework [10].

To isolate the effect of information relevance, three experimental conditions were tested: a baseline with no news access, a control where agents received random irrelevant text, and a full RAG condition with semantically matched retrieval. Each condition was run across three random seeds, and results were evaluated against the stylised fact benchmarks defined by Cont [3].

The results were unexpected. Rather than improving market realism, the RAG-informed agents produced less realistic dynamics than the random news control. The agents proved too rational holding their position in over 80% of time steps which suppressed the volatility that generates fat-tailed returns. This finding points to a tension between agent intelligence and the behavioural noise that real markets depend on.

The remainder of this paper is structured as follows. Section II reviews related work across ABM finance, RAG systems, and LLM-powered agents. Section III details the methodology,

including agent design, data corpus, and evaluation metrics. Section IV presents experimental results, and Section V discusses their implications alongside limitations and directions for future research.

II. RELATED WORK

This section reviews three bodies of literature that intersect in the present work: agent-based financial simulation, retrieval-augmented generation, and LLM-powered agents applied to financial domains. Each area has seen significant progress independently, but their intersection remains largely unexplored.

A. Agent-Based Financial Market Simulation

The idea of modelling financial markets as systems of interacting autonomous agents dates back to the late 1990s. LeBaron, Arthur, and Palmer built the Santa Fe Artificial Stock Market, one of the earliest computational models where heterogeneous agents with adaptive strategies traded a single risky asset [2]. Their work showed that simple agents, each following slightly different rules, could collectively produce price dynamics that looked surprisingly like real market data. This was a significant departure from the representative-agent models that dominated economics at the time.

Shortly after, Cont published what became a foundational reference for evaluating any financial market model [3]. He catalogued a set of statistical regularities termed stylised facts that appear consistently across different markets, time periods, and asset classes. The most prominent of these include excess kurtosis in return distributions (values well above 3.0, indicating fat tails), volatility clustering (large price changes tend to follow large price changes), absence of linear autocorrelation in raw returns, and non-stationarity of price levels. Any simulation claiming to produce realistic market behaviour needs to reproduce at least some of these properties. The present work uses Cont’s benchmarks as its primary evaluation criteria.

Market microstructure also matters. Gould et al. provided a thorough survey of limit order book models [11], explaining how the mechanics of order matching influence price formation. While a full limit order book is beyond the scope of this project, the price-impact model used here draws on the same intuition: prices move in response to the net imbalance between buy and sell pressure.

Farmer, Shubik, and Smith argued that economic systems could be studied with the same rigour as physical systems, advocating for computational modelling grounded in empirical data [4]. This perspective motivates the experimental approach taken here treating the simulation as a controlled experiment and measuring its output against known statistical properties.

More recently, Axtell and Farmer published a comprehensive survey covering two decades of ABM research in economics and finance [1]. They note that while ABMs have matured considerably, most still lack mechanisms for agents to process unstructured information like news or reports. This observation directly motivates the current work. In a related vein, Liang et al. reviewed multi-agent reinforcement learning

algorithms [12], highlighting the growing interest in systems where multiple autonomous agents learn and adapt simultaneously. Their review covers cooperative and competitive settings, though financial market simulation receives limited attention a gap this project partially addresses.

B. Retrieval-Augmented Generation

The Transformer architecture introduced by Vaswani et al. [13] underpins virtually all modern large language models. Its self-attention mechanism allows models to process long input sequences effectively, which is essential for tasks that require reasoning over multiple pieces of text. However, even large Transformer-based models suffer from a core limitation: their knowledge is frozen at training time. They cannot access new information or verify facts against external sources without additional mechanisms.

Lewis et al. proposed Retrieval-Augmented Generation (RAG) as a solution to this problem [6]. In a RAG system, a retriever component first fetches relevant documents from an external knowledge base, and these documents are then passed as context to a generative model. The generator can ground its output in the retrieved content rather than relying entirely on its parametric memory. Lewis et al. showed that this approach significantly improved performance on knowledge-intensive tasks like open-domain question answering. The architecture is straightforward to implement and has since been adopted widely in both research and industry.

A critical component of any RAG pipeline is the embedding model used to represent documents and queries as dense vectors. Reimers and Gurevych introduced Sentence-BERT [9], which adapts the BERT architecture to produce semantically meaningful sentence-level embeddings. These embeddings enable efficient similarity search given a query, the system can quickly find the most relevant documents in a large corpus. The present work uses their all-MiniLM-L6-v2 model to embed 23,515 financial news items into a FAISS index, allowing RAG agents to retrieve contextually appropriate news at each simulation step.

C. LLM-Powered Agents and Financial NLP

The use of LLMs as the cognitive backbone of autonomous agents is a relatively new but rapidly growing area. Park et al. created a simulated town populated by 25 generative agents, each driven by an LLM with access to memory and reflection modules [5]. These agents exhibited emergent social behaviours forming relationships, spreading information, and coordinating activities none of which were explicitly programmed. Their work demonstrated that LLM-powered agents can act in contextually rich, believable ways when given appropriate information retrieval mechanisms. This idea transfers naturally to financial markets: if agents can retrieve and reason about news, their trading behaviour should be more informed and potentially more realistic.

In the financial domain specifically, Yang et al. developed FinGPT, an open-source framework for building financial LLM applications [8]. FinGPT provides tools for sentiment

analysis, forecasting, and other tasks using instruction-tuned language models. Their sentiment training dataset, which aggregates labelled financial text from multiple sources, forms part of the news corpus used in this project.

Iaroshev et al. explored the application of RAG to financial report analysis [7]. They evaluated several RAG configurations for answering questions about corporate filings and found that retrieval quality had a measurable impact on answer accuracy. Their work confirms that RAG is a viable approach in financial contexts, though they focused on static question answering rather than dynamic agent decision-making.

The relationship between news sentiment and stock prices has been studied extensively through traditional methods as well. Ray, Ganguli, and Chakrabarti combined Bayesian structural time series with LSTM networks to model how news sentiment influences short-term stock price movements [14]. Their hybrid approach demonstrated that incorporating sentiment signals improves forecast accuracy, providing empirical justification for giving simulated agents access to news data.

On the data side, Malo et al. created the Financial PhraseBank [15], a corpus of nearly 5,000 sentences from financial news annotated for sentiment by domain experts. It has become a standard benchmark in financial NLP research and is one of three datasets used to build the news corpus in this project.

D. Research Gap

Each of these threads ABM finance, RAG, and LLM agents has progressed substantially, but they have developed in isolation. Agent-based financial models typically lack any form of information processing [1]. RAG systems have been tested on static QA tasks [6], [7] but not embedded within dynamic, multi-step simulations. And while LLM-powered agents have shown impressive emergent behaviour in social settings [5], they have not been deployed as traders in a market simulation and evaluated against the stylised facts that define realistic price dynamics [3].

The Mesa framework, now in version 3.x [10] and building on the original implementation by Kazil et al. [16], provides the infrastructure needed to bridge these areas. Mesa supports heterogeneous agent types, flexible scheduling through its AgentSet API, and straightforward data collection making it a natural fit for a simulation that mixes traditional rule-based agents with LLM-driven ones.

This project sits at the intersection of all three fields. It is, to the best of our knowledge, the first to embed RAG-enhanced LLM agents into a multi-agent financial market simulation and evaluate the resulting market dynamics against Cont’s stylised facts [3]. The experimental design includes a random-news control condition that isolates the effect of retrieval relevance, providing a more rigorous test than simply comparing informed agents against uninformed ones.

III. METHODOLOGY

This section describes the architecture of the simulation system, the design of each agent type, the data corpus and

retrieval pipeline, the price formation mechanism, the experimental conditions, and the metrics used for evaluation.

A. System Architecture

The simulation consists of three interconnected components: a multi-agent trading environment, a RAG pipeline for information retrieval and reasoning, and a price formation engine. The agent layer is built on Mesa 3.x [10], which provides agent management through its AgentSet API without requiring legacy scheduler objects. Mesa was chosen over alternatives like NetLogo for its native Python integration and access to the scientific computing ecosystem [16]. The RAG pipeline pairs a local FAISS vector index with Google’s Gemini 2.5 Flash-Lite model for reasoning. FAISS handles similarity search over the news corpus, while Gemini interprets the retrieved items and outputs structured trading decisions. The price formation engine sits between agent decisions and the next time step, translating aggregate buy and sell pressure into a new market price.

At each simulation step, agents observe the current price and recent price history, make a trading decision, and submit it to the order book. The order book computes a new price based on net demand, and agents then execute their trades subject to cash and share constraints. This cycle repeats for 200 steps per run.

B. Agent Design

The simulation uses 30 agents drawn from four types. Each type implements a different decision-making strategy, creating the heterogeneity that is essential for producing non-trivial market dynamics [1], [3].

FundamentalistAgent: Each fundamentalist holds a personal estimate of the stock’s true value, drawn from $\mathcal{N}(100, 5)$ at initialisation. At every step, the agent compares the current market price P_t to its fundamental value V_i . If the stock appears undervalued by more than 2%, the agent buys; if overvalued by more than 2%, it sells; otherwise it holds. Formally:

$$a_i = \begin{cases} \text{BUY} & \text{if } (V_i - P_t)/P_t > 0.02 \\ \text{SELL} & \text{if } (V_i - P_t)/P_t < -0.02 \\ \text{HOLD} & \text{otherwise} \end{cases} \quad (1)$$

These agents anchor prices around the fundamental range and are not exposed to any news data.

SentimentMomentumAgent: This agent follows short-term price momentum without any LLM involvement. At each step, it examines the last five prices to determine the prevailing trend direction. If prices have been consistently rising, the agent buys; if falling, it sells. The trade quantity reflects trend conviction: a strong, unambiguous trend triggers a larger order of 3 shares, while a weak or mixed trend produces a cautious order of 1 share. When no clear directional signal is present, the agent holds. The agent’s decision logic includes a keyword-based sentiment module capable of scanning news text for positive terms (e.g. *rise*, *gain*, *profit*) and negative terms (e.g.

fall, drop, loss), though in the current implementation the agent operates on price data alone. This design positions it as a mid-complexity benchmark more reactive than the fundamentalist, which anchors to a fixed value estimate, but lacking the information retrieval and reasoning capabilities of the RAG agent.

NoiseTrader: The noise trader selects uniformly at random from BUY, SELL, or HOLD at each step, with a small random perturbation added to the price signal. Noise traders are a standard component in ABM finance literature [2], [3]. They ensure that the market does not converge to a static equilibrium and provide the stochastic variation that can generate fat tails and volatility clustering. In the baseline condition, 10 noise traders fill the slots that RAG agents occupy in the other conditions.

RAGAgent: The RAG agent is the primary experimental subject. At each step, it constructs a text query from the current price level and recent price trajectory, then retrieves the three most semantically similar news items from the FAISS index. These items, along with price context, are formatted into a structured prompt and sent to Gemini 2.5 Flash-Lite with JSON response mode enabled. The model returns a decision in the format:

$$\{\text{action}, \text{quantity}, \text{confidence}, \text{reasoning}\} \quad (2)$$

A confidence threshold of 0.5 is applied: if the model reports confidence below this value, the action is overridden to HOLD regardless of the suggested trade. This filtering mechanism prevents the agent from acting on ambiguous or contradictory signals. An LLM response cache stores prompt-response pairs using SHA-256 hashing, so identical prompts do not consume additional API calls. Every decision is logged with its step number, action, confidence, reasoning, and the news items used enabling post-hoc qualitative analysis.

C. Data Corpus

The news corpus was assembled from three publicly available datasets. The Financial PhraseBank [15] contributed 1,584 expert-annotated sentences from financial news, providing a high-quality sentiment benchmark. From the FinGPT sentiment training set [8], 10,000 items were randomly sampled from the full 76,772 available. The Twitter Financial News Sentiment dataset added 11,931 tweets labelled as bullish, bearish, or neutral. All labels were normalised to a common three-class scheme: positive, negative, and neutral. The merged corpus totals 23,515 items.

Each text was embedded using the all-MiniLM-L6-v2 sentence transformer [9], producing 384-dimensional dense vectors. These were L2-normalised and stored in a FAISS IndexFlatIP index for cosine similarity search. The embedding and indexing process runs locally on CPU and completes in under four minutes, avoiding any dependency on external embedding APIs. At query time, retrieval of the top three items takes less than 10 milliseconds.

D. Price Formation

Prices are updated at each step using a simple price-impact model, consistent with standard ABM practice [1], [11]. After all agents submit their decisions, the order book computes the net demand:

$$\Delta = \frac{\sum q_{\text{buy}} - \sum q_{\text{sell}}}{N_{\text{agents}}} \cdot \lambda \quad (3)$$

where q_{buy} and q_{sell} are the total quantities on each side, $N_{\text{agents}} = 30$, and $\lambda = 0.05$ is a price sensitivity parameter. A Gaussian noise term $\epsilon \sim \mathcal{N}(0, 0.002)$ is added to prevent deterministic price paths. The new price is:

$$P_{t+1} = \max(0.01, P_t \cdot (1 + \Delta + \epsilon)) \quad (4)$$

This mechanism captures the basic intuition that excess buying pushes prices up and excess selling pushes them down, without the complexity of a full limit order book.

E. Experimental Design

Three conditions were tested, each run with three random seeds (42, 123, 999) for a total of nine runs. Every run lasted 200 simulation steps. The conditions differ only in the composition of the final 10 agent slots:

- **Baseline:** 10 Fundamentalist + 10 SentimentMomentum + 10 NoiseTrader. No LLM or retrieval involved.
- **Random News:** 10 Fundamentalist + 10 SentimentMomentum + 10 RAGAgent. The RAG agents receive text from a random non-financial sentence pool instead of FAISS-retrieved news.
- **Full RAG:** 10 Fundamentalist + 10 SentimentMomentum + 10 RAGAgent. The RAG agents retrieve semantically relevant news from the FAISS index.

The random news condition acts as a control. If RAG agents outperform this condition, the improvement can be attributed to retrieval relevance rather than the mere presence of text input to the LLM.

F. Evaluation Metrics

Results are evaluated against the stylised facts documented by Cont [3]. Five metrics are computed from the log returns $r_t = \ln(P_t/P_{t-1})$ of each run and averaged across seeds within each condition:

- **Excess kurtosis** > 0 : indicates fat-tailed return distributions relative to a Gaussian baseline. Computed using Fisher’s definition via `scipy.stats.kurtosis()`, where a normal distribution yields 0 and raw (Pearson) kurtosis of 3. Higher values indicate more pronounced fat tails; empirical financial returns typically exhibit excess kurtosis in the range of 3–10 [3].
- **ACF of squared returns at lag-1** > 0.10 : captures volatility clustering.
- **ACF of absolute returns at lag-1** > 0.10 : an alternative volatility clustering measure.
- **ADF test p-value** > 0.05 : failure to reject non-stationarity, consistent with random walk prices.

- **Jarque-Bera p-value** < 0.05 : rejection of normality in the return distribution.

G. Limitations of Methodology

Several constraints bound the scope of these experiments. The 200-step simulation length is shorter than the 1,000–10,000 steps typical in ABM finance studies [1], a limitation imposed by API rate constraints during development. Three random seeds provide a basic check on reproducibility but fall short of the five to ten seeds that would allow robust statistical inference. The price-impact model, while standard, omits the bid-ask spread and order queue dynamics of a full limit order book [11]. Finally, the news corpus spans 2013 to 2024 and does not include post-2024 market events, though the simulation tests the RAG architecture itself rather than the predictive value of any particular headline.

IV. RESULTS

This section presents the outcomes of the nine experimental runs across three conditions. Results are reported as averages over three random seeds (42, 123, 999) per condition unless otherwise noted. All metrics are computed from log returns $r_t = \ln(P_t/P_{t-1})$.

A. Price Dynamics

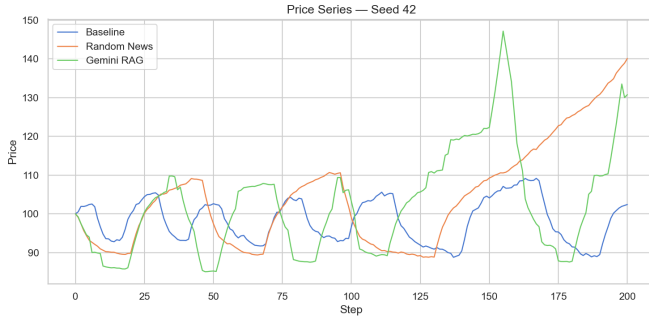


Fig. 1: Price trajectories for seed 42 across three experimental conditions.

Fig. 1 shows the price trajectories for seed 42 across all three conditions. The baseline condition stays within a relatively narrow band between 90 and 110, reflecting the stabilising effect of fundamentalist agents pulling the price toward 100. The random news condition exhibits wider swings, with prices ranging from roughly 88 to 140 by the end of the run. The full RAG condition shows the most pronounced single spike reaching 147 around step 150 but also periods of sustained flatness where prices barely move for 20 or more consecutive steps. These flat stretches correspond to periods where the RAG agents predominantly held their positions.

B. Return Distributions

Fig. 2 displays the return distributions pooled across all seeds for each condition, overlaid with a fitted normal curve. The baseline distribution is moderately peaked with tails that

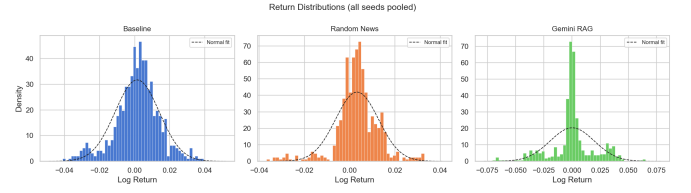


Fig. 2: Return distributions pooled across all seeds with normal fit overlay.

extend to about ± 0.04 . The random news condition produces a sharper central peak with noticeably heavier tails returns reaching ± 0.04 occur more frequently than the normal fit would predict. The full RAG condition is the most extreme in shape: a very tall, narrow spike centred at zero with occasional large outliers out to ± 0.075 . This pattern reflects an agent population that mostly holds (producing near-zero returns) but occasionally triggers sharp price moves when several RAG agents act together.

C. Stylised Fact Metrics

Table I summarises the five evaluation metrics averaged across seeds. The target values are drawn from the stylized facts documented by Cont [3].

TABLE I: Stylised fact metrics by condition. Bold = target met. Excess kurtosis uses Fisher’s definition (normal = 0; raw kurtosis = excess + 3).

Metric	Target	Baseline	Random News	Gemini RAG
Excess Kurt.	> 0	1.054	3.456	1.949
ACF(r^2) lag-1	> 0.10	0.487	0.618	0.259
ACF($ r $) lag-1	> 0.10	0.508	0.704	0.393
ADF p -value	> 0.05	0.347	0.968	0.045
JB p -value	< 0.05	0.012	0.000	0.006

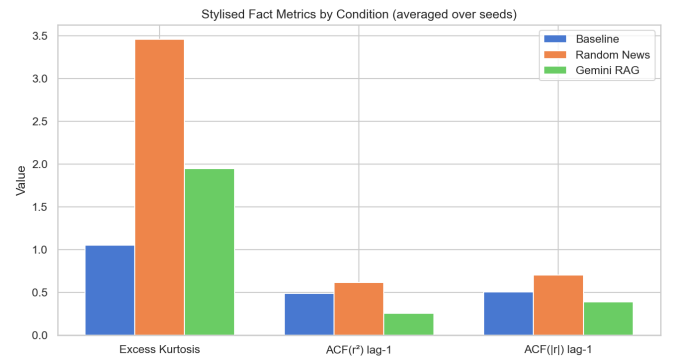


Fig. 3: Stylised fact metrics averaged over seeds. Random News exhibits the strongest excess kurtosis across all conditions.

All three conditions exhibit positive excess kurtosis, confirming fat-tailed return distributions across the board a key property of real financial markets where raw kurtosis exceeds the Gaussian value of 3.0 [3]. However, the degree of departure

varies substantially. The random news condition shows the strongest leptokurtosis at 3.456 (equivalent to a raw kurtosis of 6.456), followed by full RAG at 1.949 (raw 4.949) and baseline at 1.054 (raw 4.054). The random news condition’s excess kurtosis is roughly three times that of the baseline, indicating considerably more frequent extreme price movements. All three conditions pass the volatility clustering tests: ACF of both squared and absolute returns comfortably exceed 0.10 in every case, with the random news condition posting the strongest values (0.618 and 0.704 respectively). The full RAG condition still shows clustering, but at reduced levels (0.259 and 0.393).

The ADF test reveals a notable difference. Both baseline ($p = 0.347$) and random news ($p = 0.968$) fail to reject the unit root null hypothesis, consistent with non-stationary, random-walk-like price behaviour a key stylised fact. The full RAG condition, however, produces a borderline result ($p = 0.045$), just below the 0.05 threshold, suggesting its prices may exhibit mild mean-reversion rather than a true random walk. All three conditions reject normality under the Jarque-Bera test ($p < 0.05$), confirming non-Gaussian return distributions across the board.

Fig. 3 provides a visual comparison of the kurtosis and ACF metrics. The dominance of the random news condition on kurtosis is immediately apparent.

D. RAG Agent Behaviour

Analysis of the RAG agent decision logs from the full RAG condition (seed 42, 10 agents $\times$ 200 steps = 2,000 decisions) reveals a striking pattern. Agents chose to HOLD in 1,610 out of 2,000 decisions (80.5%), with BUY and SELL actions accounting for just 9.5% and 10.0% respectively. The confidence threshold of 0.5 was the primary driver: when retrieved news contained mixed or ambiguous signals, the model reported low confidence and the agent defaulted to holding.

Table II shows three representative decisions from different points in the simulation.

TABLE II: Sample RAG agent trading decisions (seed 42).

Step	News Retrieved	Action	Qty	Conf.	Reasoning
34	“Stocks up, trade headlines heat up”	BUY	2	0.7	Broad upward trend with positive news
6	“Markets trending lower despite positive jobs”	SELL	2	0.8	Downward trend, bearish sentiment
1	“Markets lower despite positive jobs... record territory”	HOLD	1	0.3	Mixed signals, no clear direction

The decisions show a coherent reasoning pattern. When news and price trend aligned, the agent acted with moderate-to-high confidence. When signals conflicted as in step 1, where markets were simultaneously at record levels and trending

lower the agent correctly identified the ambiguity and refrained from trading. This rational caution, replicated across most time steps, explains the dominance of HOLD actions and the resulting suppression of volatility in the full RAG condition.

E. Summary of Findings

The random news condition produced the most realistic market dynamics as measured against Cont’s stylised facts [3], meeting all five targets. The full RAG condition met four of five, producing a borderline ADF result. The baseline also met four of five, falling short only on the ADF test. The most striking outcome is that providing agents with *relevant* information made the market less realistic than providing them with *irrelevant* information a result driven by the RAG agents’ tendency to hold when faced with genuinely informative but often ambiguous news.

V. DISCUSSION AND CONCLUSIONS

A. Discussion

The central hypothesis behind this project was straightforward: agents that retrieve relevant financial news and reason about it should produce more realistic market behavior than agents operating without information. The results did not support this hypothesis. Instead, the random news condition where agents received irrelevant, non-financial text produced the most realistic stylized facts across nearly every metric. This outcome, while initially surprising, turns out to be quite explainable and arguably more interesting than a simple confirmation would have been.

The key mechanism is the confidence filter. When RAG agents received genuinely relevant financial news, the retrieved items often contained mixed or contradictory signals. A headline about strong earnings alongside one about slowing economic growth, for instance, would lead Gemini to report low confidence, and the agent would default to holding. This happened in 80.5% of all RAG agent decisions. With 10 out of 30 agents effectively sitting idle for most of the simulation, the market lost a significant source of trading activity. Fewer trades meant smaller net demand imbalances, which meant smaller price movements, which meant a narrower return distribution with an excess kurtosis of 1.95 still indicating fat tails (raw kurtosis of 4.95, above the Gaussian value of 3.0), but far less pronounced than the random news condition’s 3.46 [3].

The random news condition tells a different story. When agents received text like “The museum extended its exhibition on ancient civilizations,” Gemini had no coherent financial signal to work with. Its responses became erratic sometimes buying, sometimes selling, often with moderate confidence because the model tried to make sense of nonsensical input. This erratic behaviour injected noise into the market, functionally similar to what the noise traders do in the baseline condition but filtered through an LLM’s attempt at reasoning. The result was an excess kurtosis of 3.46 (raw kurtosis of 6.46), the strongest fat-tail signature across all conditions and well within the range typically observed in empirical financial returns [3].

Volatility clustering was also strongest here, with ACF values of 0.618 for squared returns and 0.704 for absolute returns.

There is an irony in this. The condition designed as a control the one we expected to perform worst actually performed best. And the condition designed as the main experimental treatment the one with relevant, carefully retrieved news performed second-worst on the metric that matters most. This pattern echoes something observed in other work on LLM agents. Park et al. found that their generative agents produced emergent social behaviours that were not straightforwardly predictable from the agents' individual rationality [5]. In our case, the emergent outcome is that collective rationality suppresses the very dynamics it was meant to enhance.

This finding has real implications for how ABM researchers think about integrating LLMs into financial simulations [1]. The instinct is to make agents smarter give them better data, better reasoning, better tools. But real financial markets are not populated exclusively by perfectly rational actors. Behavioural finance has long recognised that noise trading, overreaction, herding, and bounded rationality all contribute to the statistical properties of real price data [3]. A simulation that eliminates these behaviours in favour of measured, confident decision-making will, paradoxically, become less realistic.

One practical takeaway is that the confidence threshold needs careful calibration. A threshold of 0.5 was too aggressive for this setup. Lowering it to 0.3 or even 0.2 would allow more RAG agents to act on partial signals, increasing trade volume and potentially restoring the volatility needed for realistic fat tails. An alternative approach would be to introduce a stochastic override, where even confident HOLD decisions are randomly flipped to BUY or SELL with some small probability effectively injecting calibrated irrationality into otherwise rational agents.

The random news control condition proved its worth as an experimental design choice. Without it, we could only compare RAG agents to the baseline and might have concluded that the RAG architecture simply did not work. The control showed that the LLM was indeed responsive to its input it just responded to relevant input by being cautious, and to irrelevant input by being erratic. Both responses are internally consistent. The difference in market outcomes is a property of the system, not a failure of the model.

Several limitations qualify these findings. The simulation length of 200 steps is short by ABM standards, where runs of 1,000 to 10,000 steps are common [1]. Stylised facts like volatility clustering may require longer time horizons to stabilise. Only three random seeds were used per condition; five to ten would provide stronger statistical confidence in the averaged metrics. The price formation model is a simplified price-impact function rather than a full limit order book with bid-ask spreads and order queues [11]. The news corpus spans 2013 to 2024 and does not capture post-2024 market events, though since the experiment tests the retrieval architecture rather than the predictive power of specific headlines, this is a minor concern. Finally, the simulation covers a single asset only. Real markets involve cross-asset correlations and

portfolio effects that a multi-asset extension could begin to capture.

B. Future Work

Several directions follow naturally from these results. Running simulations for 1,000 or more steps with at least five seeds would test whether the observed patterns hold over longer horizons. Systematically varying the confidence threshold from 0.2 to 0.8 in increments would map out the relationship between agent caution and market realism, potentially identifying an optimal level. Extending the model to multiple correlated assets would introduce portfolio dynamics and cross-market effects. Connecting the retrieval pipeline to a live news feed rather than a static corpus would make the simulation more representative of real trading conditions. Comparing the performance of different LLMs such as GPT-4, Claude, or open-source alternatives like Llama would clarify whether the over-rationality effect is specific to Gemini or a general property of LLM-driven agents. Finally, replacing the price-impact model with a proper limit order book would bring the market microstructure closer to reality and allow investigation of bid-ask spread dynamics.

C. Conclusion

This project set out to explore whether giving simulated trading agents the ability to retrieve and reason about financial news would make artificial markets behave more like real ones. The results, drawn from nine runs across three conditions with three seeds each, offer preliminary insights rather than definitive answers the small sample size and short simulation length of 200 steps mean these findings should be treated as a starting point for further investigation.

Three tentative contributions emerge from this work. First, it presents what appears to be the first integration of a RAG pipeline with an agent-based financial market simulation. The architecture combining FAISS retrieval, LLM reasoning, and Mesa-based multi-agent coordination is modular and could serve as a foundation for more extensive future studies with larger sample sizes. Second, the experiments produced initial evidence that LLM-driven rationality may actually suppress the market dynamics it was intended to improve. Across the runs conducted, RAG agents tended to hold when faced with ambiguous information, which appeared to reduce volatility below levels typically associated with realistic markets. Meanwhile, agents fed irrelevant noise generated what looked like more realistic fat-tailed returns. Whether this pattern would hold across a larger number of seeds and longer simulations remains an open question. Third, the inclusion of a random news control condition proved useful in interpreting the results, as it helped separate the effect of having information from the effect of having relevant information though again, confirming this distinction would require broader experimental coverage.

Perhaps the most interesting observation and one that warrants further investigation is that the more informed agents appeared to produce a quieter market rather than a more realistic one. If this pattern holds under more rigorous conditions,

it would suggest that capturing the messiness of real markets may require not just better information retrieval, but some deliberate injection of the kind of irrational behaviour that drives real-world trading. For now, though, that remains a hypothesis rather than a conclusion.

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